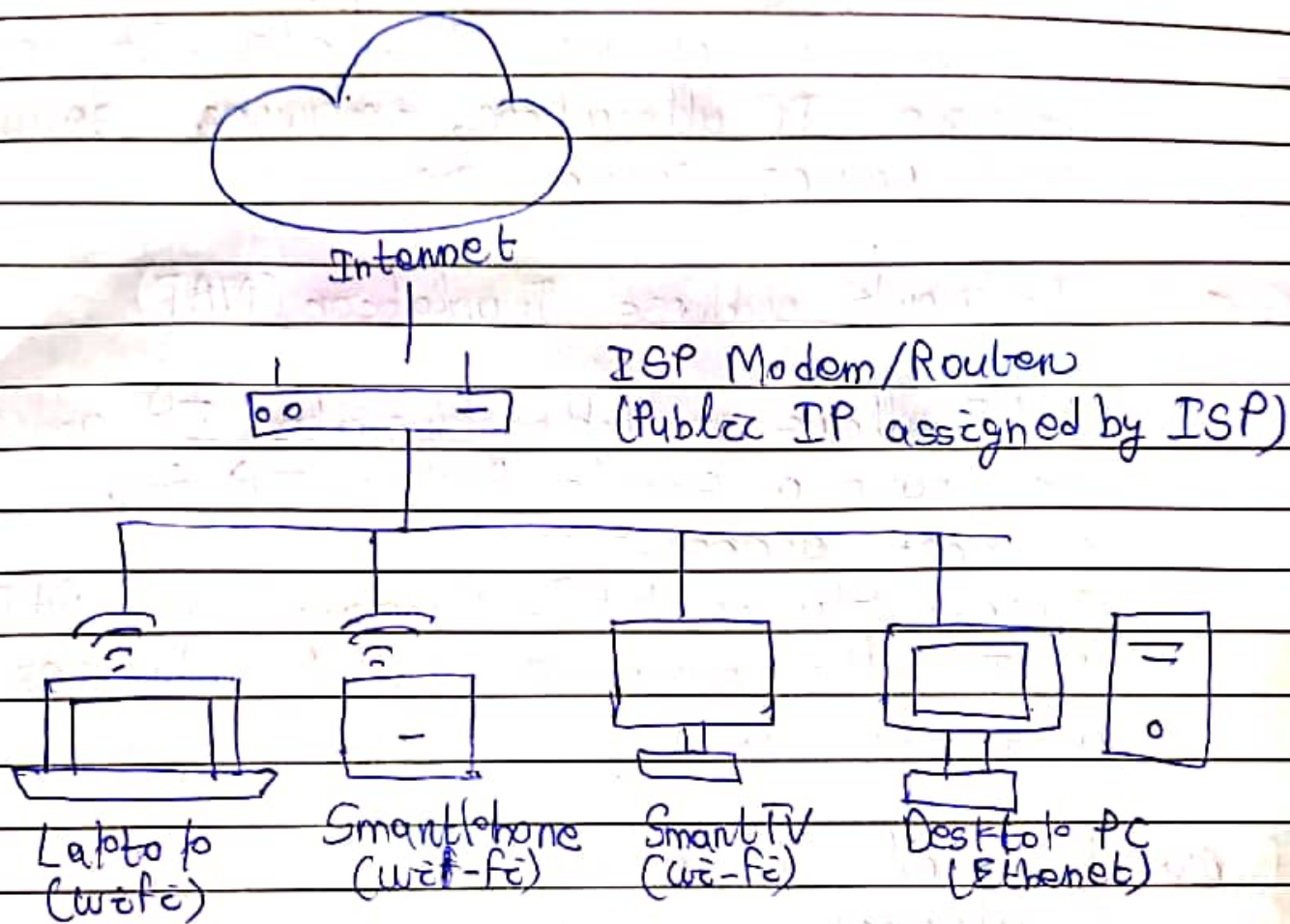


Home Network Topology

③



② IPV4 vs IPV6

- IPV4 uses 32-bit addresses, allowing ~4.3 billion addresses.
- IPV6 uses 128-bit addresses, allowing trillions of devices.
- IPV6 solves the address exhaustion problem and offers better security and routing efficiency.

② Subnetting basics:-

Subnetting divides a large network into smaller sub-networks (subnets). It helps manage IP allocation, improves security and reduces congestion.

③ Network address Translation (NAT)

NAT allows multiple private IP addresses to share a single public IP for internet access.

Types: Static NAT, Dynamic NAT, PAT
NAT helps conserve IPv4 addresses and adds a layer of security.

④ Advanced IP Address Management Strategies

- Use of DHCP for dynamic allocation
- Static IPs for critical infrastructure
- IPAM tools for central control and logging
- Role-based IP segmentation,
- Monitoring and auditing for conflicts and usage.

⑤ IP Address Allocation and Policies:

- Governed by IANA and regional RIRs. (e.g., APNIC, RIPE NCC)
- Policies ensure fair distribution and prevent abuse.
- Organizations must justify IP requests.
- Encourage IPv6 adoption.

⑥ Autonomous System (AS) Numbers:-

- An Autonomous System is a group of IP networks under one administrative domain.
- AS numbers (ASN) are used in BGP routing between networks.
- They allow better traffic control, failure, and routing optimization.

⑦ Global IP Addressing Issues & Solutions.

Issues:

- IPv4 address exhaustion
- Inefficient usage.
- Routing Scalability.

solutions:

- Transition to IPv6
- Use of NAT and CIDR
- IP reclamation and redistribution
- Promotion of IPv6-compatible hardware and software

⑧ Wireless Vs Wired Networks: Wireless (WiFi)

Factor	Wired (Ethernet)	Wireless
Speed	Faster (up to 10 Gbps)	Slower
Reliability	Very stable	Subject
Mobility	Limited	High
Security	More secure	Needs
Setup	Complex cables	Encryption
		Easy setup

⑨ Wi-Fi, Ethernet & Emerging Technologies:

- Wi-Fi 6 and 6E offer high throughput and low latency.
- Ethernet supplants higher consistent speeds (Cat6/Cat7).
- Emerging: Wi-Fi 7 (802.11 be) with higher bandwidth.
- Hybrid networks combine both technologies for flexibility.

⑩ Powerline and Optical Fiber Communication

- Powerline (PLC): Uses existing electrical wiring to transmit data; easy to install but affected by interference.
- Optical Fiber: Uses light to transmit data; offers ultra-high speed and long-distance communication; ideal for ISPs and backbone networks.

⑪ Advanced Wireless Technologies: Li-Fi

- Li-Fi: Transmits data using visible light.
- Pros: Extremely fast (Gbps), Secure (light can't pass through walls).
- Cons: Line-of-sight required, not yet widely adopted.

⑫ Performance Analysis and Network Comparison

Network	Type	Speed	Latency	Cost	Reliability
Ethernet	Very High	—	Low	Medium	High
Wi-Fi	Medium-High	—	Medium	Low	Medium
Powerline	Medium	—	Medium	Low	Medium
Optical Fiber	Ultra-High	—	Ultra low	High	Very High
Li-Fi	Very High	—	Very low	Medium	Medium